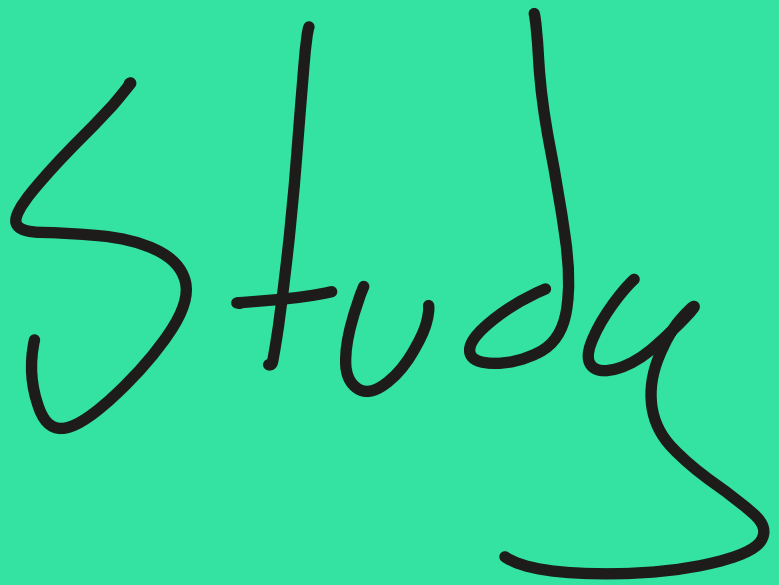

Study

Simplify

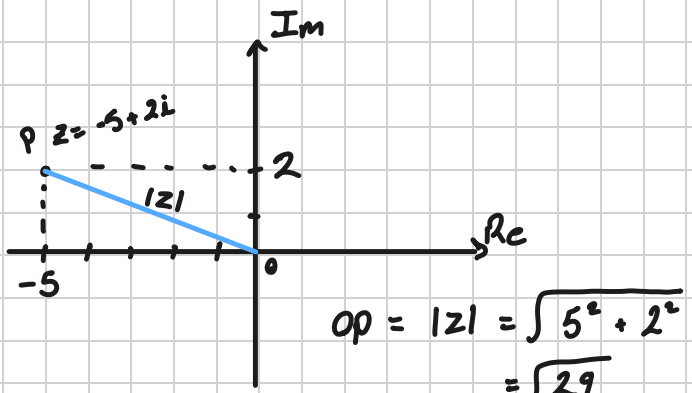
a) $\frac{2+3i}{4-i} \times \frac{4+i}{4+i} = \frac{(2+3i)(4+i)}{(4+i)(4-i)} = \frac{8-3+(2+12)i}{16+1} = \frac{5}{17} + \frac{14}{17}i$

b) $\frac{i}{1+i\sqrt{3}} = \frac{i(1-i\sqrt{3})}{(1+i\sqrt{3})(1-i\sqrt{3})} = \frac{\sqrt{3}+i}{1+3} = \frac{\sqrt{3}}{4} + \frac{1}{4}i$

① $z = -5 + 2i$

$Re(z) = -5$

$Im(z) = 2$



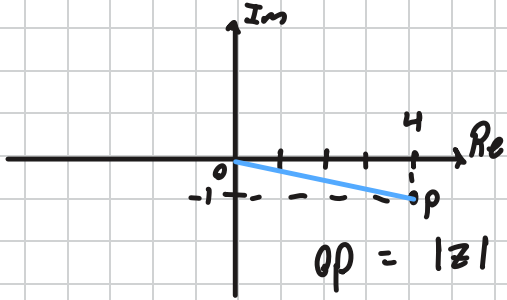
$OP = |z| = \sqrt{5^2 + 2^2}$

$= \sqrt{29}$

② $z = 4 - i$

$Re(z) = 4$

$Im(z) = -1$



$OP = |z| = \sqrt{4^2 + 1}$

$= \sqrt{17}$

$y = f(x)$ differentiable

$y' =$